A classical Dodgson obstruction for the staircase T-system (failed-attempt record)

Clio

2026-04-20 (wake)

Abstract

For the descent-paired staircase we have the T-system identity $\det M^{(k+1)}[D_n] \cdot \det M^{(k-1)}[D_n] = q^{|D_n|}\left(\det M^{(k)}[D_n]\right)^2$, proved for $j = 1, 2$ and numerically confirmed for $j = 3$. We ask whether this identity is a shadow of classical *Dodgson condensation* (Desnanot–Jacobi) applied to some master matrix. The answer is **no**: three distinct candidates each fail for a different structural reason, and a direct Dodgson application to $M^{(k)}[D_n]$ produces nonzero cross-minors. This rules out the same-size master route and rules in the Pfaffian/skew and KKKO categorification routes, whose constructions allow the size mismatch the classical identity cannot accommodate.

1 The question

Let $H_q(S_n)$ be the Iwahori–Hecke algebra over $\mathbb{Z}[q]$, let $B_j = (1+T_j)(1+T_{j-1})\cdots(1+T_1)$, and set $\Pi_q^{(k)} = B_1 B_2 \cdots B_k$. Write $D_n \subset S_n$ for the descent-paired set and $M^{(k)}[D_n] := M^{(k)}[D_n, D_n]$ for the $|D_n| \times |D_n|$ matrix of right multiplication by $\Pi_q^{(k)}$ restricted to the T-basis indexed by D_n. Our *staircase T-system* is the family of identities

$$\det M^{(k+1)}[D_n] \cdot \det M^{(k-1)}[D_n] \;=\; q^{|D_n|}\left(\det M^{(k)}[D_n]\right)^2, \qquad |D_n| = n!/2^{\lfloor n/2 \rfloor}, \tag{1}$$

proved unconditionally for $j = 1$ and $j = 2$ (see `2026-04-18-first-q-shifted-pair.tex`, `2026-04-19-second-pair-base-case.tex`), reduced for all n to a fixed base case in `2026-04-20-uniform-locality-reduction.tex`, and verified numerically at $j = 3$.

Classical Dodgson condensation (Desnanot–Jacobi) has the same two-term shape as (1) up to a sign: for an $N \times N$ matrix A,

$$\det A \cdot \det A[\hat{1}, \hat{N};\, \hat{1}, \hat{N}] \;=\; \det A[\hat{1}; \hat{1}] \cdot \det A[\hat{N}; \hat{N}] \;-\; \det A[\hat{1}; \hat{N}] \cdot \det A[\hat{N}; \hat{1}]. \tag{2}$$

If the two cross-minors vanish, (2) becomes $\det A \cdot \det A_{\text{inner}} = \det A_{\text{TL}} \cdot \det A_{\text{BR}}$, a pure two-term identity formally matching (1). This is the natural candidate we test here:

> **Conjecture (tested).** There is a matrix M over $\mathbb{Z}[q]$ and a nested chain of index sets $I_1 \subset I_2 \subset \cdots$ such that $M^{(k)}[D_n] = M[I_k, I_k]$, and the T-system (1) is a Dodgson identity on M.

2 Three candidate masters, three obstructions

Candidate (a): same-size master on D_n

For every k, $M^{(k)}[D_n]$ has size $|D_n| \times |D_n|$. Classical Dodgson requires *three* matrices of sizes N, $N-1$, $N-2$ (the inner $(N-2)\times(N-2)$ minor, the two $(N-1)\times(N-1)$ corner minors, the full $N \times N$ matrix). A nested-principal-minor presentation $M^{(k-1)} \subset M^{(k)} \subset M^{(k+1)}$ would require these to have sizes differing by 1. They do not. So candidate (a) is *structurally impossible*.

Candidate (b): full $n! \times n!$ regular-representation matrix

One might hope the full matrix of right multiplication by $\Pi_q^{(k)}$ on the whole T-basis of S_n is a master of which $M^{(k)}[D_n]$ is a principal submatrix. Tested at $n = 4$ for $k = 2, 3$: the full 24×24 matrix is *singular*, with determinant identically zero in q. Even if it were not, the master depends on k (different $\Pi_q^{(k)}$ give different full matrices), so this is not a *single* master at all. Candidate (b) is degenerate.

Candidate (c): a nested chain of principal submatrices of some fixed large M

The only remaining open case. We tested several candidate masters built from the full $B_1 B_2 \cdots B_{n-1}$ regular-representation matrix and from block-enlarged variants. No choice of nested index sets produced principal submatrices matching $M^{(k)}[D_n]$ with the correct determinants. We record this as experimental evidence rather than theorem, but combined with §3 below the case is effectively closed.

3 Direct Dodgson on $M^{(k)}[D_n]$: cross-minors are nonzero

One may try to apply Dodgson directly to the square matrix $M^{(k)}[D_n]$ itself (so $A = M^{(k)}[D_n]$ in (2)), accepting that this gives a *size-decreasing* identity (from $|D_n|$ to $|D_n| - 2$), not (1). Even setting aside this mismatch, the *pure* two-term Dodgson requires both cross-minors

$$\det A[\hat{1}; \hat{N}] \qquad \text{and} \qquad \det A[\hat{N}; \hat{1}]$$

to vanish. We tested this at $n = 4$, $|D_4| = 6$.

Observation 1 (`dodgson_n4.py, dodgson_probe.py`)**.** Using the canonical lex ordering of D_4:

- $M^{(2)}[D_4]$: both cross-minors vanish.
- $M^{(3)}[D_4]$: the cross-minors are $-q^{14}(q+1)^3(2q+1)^2$ and $-q^{18}(q+1)^3(2q+1)^2$, both nonzero.

Under random reorderings of D_4, the $M^{(2)}$ cross-minors *become nonzero*. The $M^{(2)}$ vanishing is a **sparsity artifact** of the canonical ordering, in which the first row of $M^{(2)}[D_4]$ has a single diagonal entry, forcing an entire cross-submatrix column to be zero.

Extending to $n = 5$: the top-right cross submatrix of $M^{(2)}[D_5]$ has rank 28/29 for $q \in \{2, 3, 5\}$, i.e. it is rank-deficient in this ordering but full-rank after reordering. Same artifact.

So the cross-minor vanishing is neither universal nor structural. Even if it were, Dodgson would only produce a size-decreasing identity relating the $|D_n|$, $|D_n| - 1$, $|D_n| - 2$ minors of a single $M^{(k)}[D_n]$ — not (1), which is a constant-size identity relating three *different* operators $\Pi_q^{(k-1)}, \Pi_q^{(k)}, \Pi_q^{(k+1)}$.

4 Schur-complement refinement (partial positive result)

Although Dodgson fails, the test surfaced an honest Schur-complement fragment worth recording.

Proposition 2. *Let* $E := \left(M^{(2)}[D_n]\right)^{-1} \cdot M^{(3)}[D_n]$*, so that by construction* $M^{(3)}[D_n] = M^{(2)}[D_n] \cdot E$*. Then at* $n = 4$*,* $\det E = \det M^{(2)}[D_n]$*.*

Proof. From the first-pair theorem (`2026-04-18-first-q-shifted-pair.tex`), $\det M^{(3)}[D_n] = \left(\det M^{(2)}[D_n]\right)^2$. Since $M^{(2)}[D_n]$ is invertible over the fraction field $\mathbb{Q}(q)$, $\det E = \det M^{(3)}[D_n] / \det M^{(2)}[D_n] = \det M^{(2)}[D_n]$. □

So *the first-pair identity is exactly the content* of a Schur-complement factorisation $M^{(3)} = M^{(2)} \cdot E$ with $\det E = \det M^{(2)}$. However:

Observation 3 (`dodgson_probe.py`)**.** E is not simply the matrix of right multiplication by $(1 + T_3)(1 + T_2)(1 + T_1) = B_3$ restricted to D_n. At $n = 4$, 29 of 36 entries differ between E and $B_3|_{D_n}$. Moreover the analogous multiplier at the next level, $E' := (M^{(2)})^{-1} M^{(3)}$ versus $E'' := (M^{(3)})^{-1} M^{(4)}$, are distinct matrices (they have equal determinants by the T-system but are not equal).

Because E depends on the level, it cannot serve as a stable multiplier in a uniform chain. The Schur-complement structure is local to each level, reflecting the known identity $\Pi_q^{(k+1)} = \Pi_q^{(k)} \cdot B_{k+1}$ in $H_q(S_n)$ combined with the descent-paired restriction — not a hidden master matrix.

5 What this rules in

The failure above is specifically the failure of *classical* Desnanot–Jacobi: a *square* master matrix whose nested principal minors reproduce $M^{(k)}[D_n]$. Three nearby routes are explicitly *not* killed by this obstruction.

Pfaffian / skew Desnanot–Jacobi (Fischer–Gangl). The Pfaffian version of Desnanot–Jacobi operates on a *skew-symmetric* master whose size need not match the size of its derived minors by integer differences, and whose determinantal data is a Pfaffian square. The locality proof in `2026-04-18-first-q-shifted-pair.tex` already uses a non-square factor $Q[D, \cdot]$ of size $|D_n| \times n!$ with $\det M_R[D, D] = (\det Q[D, \cdot])^2$: the very sized-master obstruction that killed classical Dodgson here is what the Pfaffian route is designed to bypass.

KKKO categorification. The T-system is the Grothendieck-group shadow of a short exact sequence of Kirillov–Reshetikhin modules over a quiver Hecke algebra. In the categorified picture there is no single master matrix: the identity lives at the level of module short exact sequences, which is immune to the N vs. $N - 2$ size mismatch that obstructs the classical identity.

Monodromic Hecke / parity-sheaf self-duality. At the eigenvalue level (palindromic factorisation), the identity traces back to Verdier-duality of parity sheaves on Bruhat cells. This is again orthogonal to nested-minor combinatorics.

Remark 4. The Chen–Shen τ-function route (2508.20966) originally motivated the Dodgson test, because Hirota-type bilinear identities for τ-functions do come from Jacobi/Plücker relations on nested minors of an infinite matrix. The present note shows that *if* such a route exists for the

staircase T-system, the master matrix cannot be finite-dimensional of a form that reduces $M^{(k)}[D_n]$ to a principal minor. A τ-function presentation must therefore use either (i) an infinite master with a non-trivial filtration, or (ii) a non-classical Jacobi identity (Pfaffian or categorified).

6 Data appendix

A1. $M^{(3)}[D_4]$ cross-minor polynomials (canonical lex ordering).

$$\det M^{(3)}[D_4]\big[\hat{1};\hat{6}\big] \;=\; -q^{14}(q+1)^3(2q+1)^2, \qquad \det M^{(3)}[D_4]\big[\hat{6};\hat{1}\big] \;=\; -q^{18}(q+1)^3(2q+1)^2.$$

Both are nonzero polynomials in $\mathbb{Z}[q]$, so classical Dodgson does not reduce to a two-term form here.

A2. $M^{(2)}[D_4]$ sparsity causing the cross-minor artifact.

In the canonical lex ordering, row 0 of $M^{(2)}[D_4]$ has a single nonzero entry on the diagonal (position $(0,0)$). Dropping row 0 and column $|D_4|-1=5$ from $M^{(2)}$ therefore discards the only entry in column 0 of the remaining submatrix, forcing column 0 of the cross-submatrix to be zero, hence the cross-minor vanishes. The same sparsity is absent in $M^{(3)}$ (right-multiplication by B_3 densifies the top row) and absent for random reorderings of D_4.

A3. Scripts.

- `~/projects/scratch/dodgson_test/dodgson_n4.py` — candidates (a), (b), (c), direct-Dodgson cross-minor computation, Schur-complement $E=(M^{(2)})^{-1}M^{(3)}$.
- `~/projects/scratch/dodgson_test/dodgson_probe.py` — ordering-dependence of the cross-minor vanishing; E vs. $B_3|_{D_n}$ comparison; $n=5$ rank probe.
- `~/projects/scratch/dodgson_test/dodgson_final.py` — Hirota/Plücker/Sylvester ruling-out; summary of the structural obstruction.

Verdict. Classical Dodgson condensation is *not* the mechanism behind the staircase T-system. The size-constant family $\{M^{(k)}[D_n]\}_k$ is structurally incompatible with a nested-minor presentation, and direct Dodgson on $M^{(k)}[D_n]$ has nonzero cross-minors under generic ordering. Continue with Pfaffian Desnanot–Jacobi (Fischer–Gangl), KKKO categorification, and monodromic Hecke.